AI EXPONENT LLC

EU AI Act Article 9 Risk Management File

Retail Credit Risk Scorer

Version 1.0.0

28	14	8
risk items	open above appetite	knowledge gaps

Provider	Northwind Scoring Labs (fictional)
Assessor	Mara Lindqvist, Head of Model Risk
Assessment date	04 November 2026
Review date	30 October 2027
Annex III category	Essential Services
Readiness gates	Exported with failing gates: G3

Content digest (SHA-256)

`9e53abfb36a7d22a8ed445f185e3ef0e8fc484d72f3a1c170845985f8af4bb23`

1. Executive summary

Purpose: Credit scoring of retail consumer credit applicants: a binary classifier, trained on the UCI Statlog German Credit data, scores each applicant as good or bad credit risk, and a lender's credit officer uses the score when deciding on the application.

Annex III category: Essential Services

Self-classification documented: Yes

Risk items: 28 in total. Before treatment, 16 scored above the risk appetite threshold of 9; after treatment, 14 remain open.

Knowledge gaps: 8

System description

Field	Value
System ID	`2c475c5c-c646-4da8-89a9-03141a0bcfbd`
Name	Retail Credit Risk Scorer
Version	1.0.0
Provider	Northwind Scoring Labs (fictional)
Provider contact	*Not specified*
Intended users	*Not specified*
Inputs	*Not specified*
Outputs	*Not specified*
Deployment context	*Not specified*

This document was produced using RiskForge v2.0.0, question bank version 1.0.0. It represents the team's documented risk assessment and has not been reviewed by a qualified legal professional. It does not constitute legal advice under the EU AI Act or any other regulation.

2. Methodology

The assessment uses the RiskForge question bank, version 1.0.0, across 8 risk dimensions: health safety, fundamental rights, discrimination, privacy, transparency, human oversight, robustness, data governance.

Each risk is scored for likelihood and severity on a five-point scale. The risk score is likelihood times severity, from 1 to 25. Bands: low 1 to 4, medium 5 to 9, high 10 to 16, critical 17 to 25. Likelihood and severity are scored again after mitigation to give the residual score.

A risk is open while its residual score is above the risk appetite threshold of 9 and it has not been accepted with a written rationale. Gate G3 fails while any risk is open.

Risk patterns applied: CREDIT_SCORING_BIAS.

Readiness gates G1 to G8 are the checks `riskforge validate` runs. A FAIL gate blocks export unless the export is forced. Results at export:

G1 PASS, **G2 PASS**, **G3 FAIL**, **G4 PASS**, **G5 PASS**, **G6 PASS**, **G7 PASS**, **G8 PASS**

3. Risk register

Risk appetite threshold: 9 (residual scores above it need mitigation or acceptance).

3.1 Health safety

Are system performance metrics (accuracy, error rates, latency) monitored in production with alerts for degradation? HIGH (12)

Post-market monitoring under Article 72 requires tracking of actual performance against stated specifications.

Likelihood	likely (4)	**Severity**	moderate (3)
Residual likelihood	likely (4)	**Residual severity**	moderate (3)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.9(8), Art.72

NIST AI RMF: MEASURE 2.4 | ISO/IEC 42001: Clause 9.1

Mitigations:

- **Detective** [planned]: Track approval rate, observed bad rate at 12 months and cost-weighted error each month against the validation values, with an alert at a 20 percent relative change.
  Recorded by: Mara Lindqvist, Head of Model Risk

Could vulnerable populations (elderly, children, people with disabilities) be disproportionately harmed by system errors... HIGH (12)

Article 9(9) requires explicit consideration of impacts on children and other vulnerable groups.

Likelihood	possible (3)	**Severity**	major (4)
Residual likelihood	possible (3)	**Residual severity**	major (4)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.9(9)

NIST AI RMF: MAP 5.1 | ISO/IEC 42001: Clause 6.1

Mitigations:

- **Detective** [planned]: Report error rates and cost-weighted error by age band (under 25, 25 to 59, 60 and over) at each release and block release if a band is worse than the overall rate by more than 10 percent.
  Recorded by: Mara Lindqvist, Head of Model Risk

3.2 Fundamental rights

Has a fundamental rights impact assessment (FRIA) been conducted prior to deployment? MEDIUM (9)

Article 9(2)(a) requires identification of known and foreseeable risks to fundamental rights. A FRIA is the standard mechanism for this.

Likelihood	possible (3)	**Severity**	moderate (3)
Residual likelihood	possible (3)	**Residual severity**	moderate (3)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.9(2)(a)

NIST AI RMF: GOVERN 1.1 | ISO/IEC 42001: Clause 6.1

Knowledge gap: its test requirements are listed in section 5.

Could the system restrict or affect access to rights protected under the EU Charter of Fundamental Rights? HIGH (12)

Consider right to an effective remedy, right to a fair trial, presumption of innocence, freedom of expression, right to education.

Likelihood	possible (3)	**Severity**	major (4)
Residual likelihood	possible (3)	**Residual severity**	major (4)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.9(2)(a)

NIST AI RMF: MAP 5.1 | ISO/IEC 42001: Clause 6.1

Mitigations:

- **Corrective** [planned]: State in the instructions for use that a refused applicant can ask for a second review by a credit officer who did not make the first decision, and that the score alone must not refuse an application.
  Recorded by: Mara Lindqvist, Head of Model Risk

Are affected individuals informed of the AI system's role in decisions that affect them, in a language they understand? MEDIUM (6)

Transparency is a foundational right. The obligation to inform subjects is strengthened in high-risk AI contexts.

Likelihood	unlikely (2)	**Severity**	moderate (3)
Residual likelihood	unlikely (2)	**Residual severity**	moderate (3)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.26(11)

NIST AI RMF: MEASURE 2.8 | ISO/IEC 42001: Annex A.6

Knowledge gap: its test requirements are listed in section 5.

3.3 Discrimination

Demographic bias in credit scoring outputs

HIGH (12)

Credit scoring models trained on historical data may encode systemic biases against protected groups (age, gender, ethnicity). Article 9(9) requires consideration of impacts on vulnerable groups.

Likelihood	possible (3)	**Severity**	major (4)
Residual likelihood	possible (3)	**Residual severity**	major (4)
Source	pattern	**Regulatory status**	settled

Article refs: Art.9(2)(a), Art.9(9), Art.10(2)(f)

NIST AI RMF: MEASURE 2.11 | ISO/IEC 42001: Annex A.7

Mitigations:

- **Detective** [planned]: Apply the release gate on demographic parity and equal opportunity difference (above 0.10 blocks release) by sex, age band and foreign worker flag.
  Recorded by: Mara Lindqvist, Head of Model Risk

Has the training dataset been audited for representation imbalances across protected characteristics (gender, ethnicity,...

HIGH (16)

Biased training data produces biased outputs. Article 10(2)(f) requires data governance measures to address biases.

Likelihood	likely (4)	**Severity**	major (4)
Residual likelihood	likely (4)	**Residual severity**	major (4)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.10(2)(f), Art.9(2)(a)

NIST AI RMF: MEASURE 2.11 | ISO/IEC 42001: Annex A.7

Mitigations:

- **Detective** [implemented]: Report record counts and the share of bad class by the sex part of attribute 9, by age band (under 25, 25 to 59, 60 and over) and by foreign worker flag, and state that married and divorced women share one code.
  Recorded by: Mara Lindqvist, Head of Model Risk

Have demographic parity, equalised odds, or other fairness metrics been computed and documented for each protected group... HIGH (16)

Quantified fairness metrics are required for meaningful bias assessment. Select metrics appropriate to the use case.

Likelihood	likely (4)	**Severity**	major (4)
Residual likelihood	likely (4)	**Residual severity**	major (4)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.9(8), Art.10(2)(f)

NIST AI RMF: MEASURE 2.11 | ISO/IEC 42001: Annex A.7

Mitigations:

- **Detective** [planned]: Compute demographic parity difference and equal opportunity difference by sex, age band and foreign worker flag on the hold-out set; block release if any difference is above 0.10.
  Recorded by: Mara Lindqvist, Head of Model Risk

Does the system use proxy variables that may correlate with protected characteristics (e.g. postcode, school name, emplo... CRITICAL (20)

Proxy discrimination occurs when facially neutral features encode protected characteristics. Legal exposure exists even without intent.

Likelihood	almost_certain (5)	**Severity**	major (4)
Residual likelihood	possible (3)	**Residual severity**	major (4)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.9(2)(a), Art.10(2)(f)

NIST AI RMF: MEASURE 2.11 | ISO/IEC 42001: Annex A.7

Mitigations:

- **Preventive** [implemented]: Remove attributes 9 (personal status and sex), 13 (age) and 20 (foreign worker) from the model inputs. Keep them in a separate table used only for bias testing.
  Recorded by: Mara Lindqvist, Head of Model Risk
- **Detective** [planned]: Measure the association of each remaining input with sex, age band and foreign worker status on the 1000 records, and drop or bin any input with Cramer's V above 0.3.
  Recorded by: Mara Lindqvist, Head of Model Risk

Are bias test results documented and reviewed at each model update, not just at initial deployment? MEDIUM (6)

Data drift and concept drift can re-introduce bias after initial mitigation. Continuous monitoring is required.

Likelihood	unlikely (2)	**Severity**	moderate (3)
Residual likelihood	unlikely (2)	**Residual severity**	moderate (3)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.9(8), Art.72

NIST AI RMF: MEASURE 2.11 | ISO/IEC 42001: Clause 9.1

Knowledge gap: its test requirements are listed in section 5.

3.4 Privacy

Does the system process personal data or special category data (health, biometric, ethnicity, religion)? CRITICAL (20)

Processing of special category data triggers additional obligations under GDPR Article 9. EU AI Act Article 4a(1) allows providers of high-risk AI systems to process special categories of personal data exceptionally, to the extent strictly necessary to ensure bias detection and correction.

Likelihood	almost_certain (5)	**Severity**	major (4)
Residual likelihood	almost_certain (5)	**Residual severity**	major (4)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.4a(1)

NIST AI RMF: MEASURE 2.10 | ISO/IEC 42001: Annex A.7

Mitigations:

- **Preventive** [planned]: Record the legal basis for using the 1994 records and live applicant data, agreed with the lender's data protection officer, and list the personal data each input carries.
  Recorded by: Mara Lindqvist, Head of Model Risk

Has a Data Protection Impact Assessment (DPIA) been carried out, with the advice of the Data Protection Officer where on... MEDIUM (9)

GDPR Article 35 requires a DPIA for high-risk processing. High-risk AI systems typically trigger this threshold.

Likelihood	possible (3)	**Severity**	moderate (3)
Residual likelihood	possible (3)	**Residual severity**	moderate (3)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.26(9), Art.9(2)(a)

NIST AI RMF: MEASURE 2.10 | ISO/IEC 42001: Annex A.7

Knowledge gap: its test requirements are listed in section 5.

Is data minimisation enforced: does the model use only the minimum personal data necessary for the intended purpose? HIGH (12)

GDPR data minimisation and purpose limitation principles (GDPR Article 5(1)(b) and (c)) apply to AI systems processing personal data.

Likelihood	likely (4)	**Severity**	moderate (3)
Residual likelihood	unlikely (2)	**Residual severity**	moderate (3)
Source	question_bank	**Regulatory status**	settled

NIST AI RMF: MEASURE 2.10 | ISO/IEC 42001: Annex A.7

Mitigations:

- **Preventive** [implemented]: Remove attribute 19 (telephone) as well as attributes 9, 13 and 20 from the model inputs, and document why each remaining input is needed.
  Recorded by: Mara Lindqvist, Head of Model Risk

Has the training data been scanned for inadvertent PII inclusion, and is PII redaction/pseudonymisation applied before t... MEDIUM (6)

Training data containing real PII creates privacy risk. Automated PII detection tools (e.g. TraceForge) can surface this.

Likelihood	unlikely (2)	**Severity**	moderate (3)
Residual likelihood	unlikely (2)	**Residual severity**	moderate (3)
Source	question_bank	**Regulatory status**	settled

NIST AI RMF: MEASURE 2.10 | ISO/IEC 42001: Annex A.7

3.5 Transparency

Lack of explanation for credit decisions

MEDIUM (9)

Where a solely automated credit decision falls under GDPR Article 22(2)(a) or (c), GDPR Article 22(3) gives at least the right to obtain human intervention, to express a point of view and to contest the decision, and GDPR Articles 13(2)(f), 14(2)(g) and 15(1)(h) require meaningful information about the logic involved. The system may not currently support this.

Likelihood	possible (3)	**Severity**	moderate (3)
Residual likelihood	possible (3)	**Residual severity**	moderate (3)
Source	pattern	**Regulatory status**	settled

Article refs: Art.13, Art.86(1)

NIST AI RMF: MEASURE 2.8 | ISO/IEC 42001: Annex A.6

Mitigations:

- **Preventive** [planned]: Return the four inputs that contributed most to each score as reason codes, for the lender to pass to the applicant on request.
  Recorded by: Mara Lindqvist, Head of Model Risk

Are users notified that they are interacting with or being assessed by an AI system?

MEDIUM (9)

Article 26(11) requires deployers of Annex III high-risk AI systems that make or assist in making decisions related to natural persons to inform them that they are subject to the use of the system. Article 50(1) covers informing natural persons that they are interacting with an AI system.

Likelihood	possible (3)	**Severity**	moderate (3)
Residual likelihood	possible (3)	**Residual severity**	moderate (3)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.26(11), Art.50(1)

NIST AI RMF: MEASURE 2.8 | ISO/IEC 42001: Annex A.6

Knowledge gap: its test requirements are listed in section 5.

Is an explanation of the system's decision-making logic available to affected individuals upon request?

HIGH (12)

GDPR Article 22(1) gives a right not to be subject to a decision based solely on automated processing, including profiling, which produces legal effects or similarly significantly affects the person. GDPR Articles 13(2)(f), 14(2)(g) and 15(1)(h) require meaningful information about the logic involved. An explanation capability is a key technical control.

Likelihood	possible (3)	**Severity**	major (4)
Residual likelihood	possible (3)	**Residual severity**	major (4)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.13(3)(b)(iv), Art.86(1)

NIST AI RMF: MEASURE 2.8 | ISO/IEC 42001: Annex A.6

Mitigations:

- **Preventive** [planned]: Return the four inputs that contributed most to each score as reason codes, for the lender to pass to the applicant on request.
  Recorded by: Mara Lindqvist, Head of Model Risk

Does the system documentation include clear capability limitations, known failure modes, and out-of-distribution behavio...

MEDIUM (9)

Article 13(3)(b) requires information about the system's capabilities and limitations for the benefit of deployers.

Likelihood	possible (3)	**Severity**	moderate (3)
Residual likelihood	unlikely (2)	**Residual severity**	moderate (3)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.13(3)(b)

NIST AI RMF: MEASURE 2.8 | ISO/IEC 42001: Annex A.6

Mitigations:

- **Preventive** [implemented]: Add a limitations section to the instructions for use: the training data is from 1994 and in DM, the collection and the meaning of good and bad are not documented, representativeness is unknown, and code A47 is uncertain.
  Recorded by: Mara Lindqvist, Head of Model Risk

3.6 Human oversight

Are human reviewers trained on the system's capabilities, limitations, and known biases before making decisions based on... MEDIUM (9)

Untrained human reviewers provide illusory oversight. Article 26(2) requires deployers to assign human oversight to natural persons who have the necessary competence, training and authority.

Likelihood	possible (3)	**Severity**	moderate (3)
Residual likelihood	possible (3)	**Residual severity**	moderate (3)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.26(2)

NIST AI RMF: MAP 3.5 | ISO/IEC 42001: Annex A.9

Knowledge gap: its test requirements are listed in section 5.

Is the human oversight load (volume of cases reviewed per reviewer per day) realistic for meaningful review, not rubber-... HIGH (12)

Automation bias and review fatigue cause human oversight to become nominal. Workload design is a technical control.

Likelihood	possible (3)	**Severity**	major (4)
Residual likelihood	possible (3)	**Residual severity**	major (4)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.14

NIST AI RMF: MAP 3.5 | ISO/IEC 42001: Annex A.9

Knowledge gap: its test requirements are listed in section 5.

Is the system designed to make its reasoning interpretable to human reviewers (e.g. via explainability methods, confiden... HIGH (12)

Human oversight is only meaningful if the reviewer can understand what the system decided and why.

Likelihood	possible (3)	**Severity**	major (4)
Residual likelihood	possible (3)	**Residual severity**	major (4)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.14(4)(c)

NIST AI RMF: MEASURE 2.9 | ISO/IEC 42001: Annex A.6

Mitigations:

- **Preventive** [planned]: Show the credit officer the reason codes and the score band next to each score.
  Recorded by: Mara Lindqvist, Head of Model Risk

3.7 Robustness

Has the system been tested against out-of-distribution inputs, edge cases, and adversarial perturbations?

HIGH (16)

Article 9(6) and (8) require testing against prior defined metrics and probabilistic thresholds appropriate to the intended purpose. Article 15(5) covers resilience against adversarial examples.

Likelihood	likely (4)	**Severity**	major (4)
Residual likelihood	likely (4)	**Residual severity**	major (4)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.9(6), Art.15

NIST AI RMF: MEASURE 2.5 | ISO/IEC 42001: Annex A.9

Mitigations:

- **Detective** [planned]: Before release, test on a recent sample of the lender's own applicants with amounts converted from DM to euro, plus edge cases (extreme amount and duration) and small input perturbations; the cost-weighted error must not exceed the hold-out value by more than 10 percent.
  Recorded by: Mara Lindqvist, Head of Model Risk

Are accuracy, precision, recall, and calibration metrics documented across all relevant demographic subgroups?

HIGH (16)

Aggregate metrics can mask poor performance for subpopulations. Slice-level evaluation is required for high-risk systems.

Likelihood	likely (4)	**Severity**	major (4)
Residual likelihood	likely (4)	**Residual severity**	major (4)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.9(8), Art.10(2)(f)

NIST AI RMF: MEASURE 2.5 | ISO/IEC 42001: Annex A.9

Mitigations:

- **Detective** [planned]: Report accuracy, precision, recall, calibration error and cost under the dataset's cost matrix (5 for a bad applicant scored good, 1 for the reverse) per subgroup on the hold-out set.
  Recorded by: Mara Lindqvist, Head of Model Risk

Is there a defined performance floor (minimum acceptable metric threshold) below which the system is taken offline? MEDIUM (9)

A defined performance floor ensures degraded models do not remain in production. This is a key operational control.

Likelihood	possible (3)	**Severity**	moderate (3)
Residual likelihood	possible (3)	**Residual severity**	moderate (3)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.9(8), Art.72

NIST AI RMF: MANAGE 1.3 | ISO/IEC 42001: Clause 9.1

Are model degradation scenarios (data drift, concept drift, distribution shift) monitored and have defined response proc... CRITICAL (20)

Models degrade over time as real-world data distributions shift. Post-market monitoring under Article 72 requires detection and response.

Likelihood	almost_certain (5)	**Severity**	major (4)
Residual likelihood	almost_certain (5)	**Residual severity**	major (4)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.72

NIST AI RMF: MEASURE 2.4 | ISO/IEC 42001: Clause 9.1

Mitigations:

- **Preventive** [planned]: Retrain on recent applicant data from the lender before release, and do not put the model trained only on the 1994 records into production.
  Recorded by: Mara Lindqvist, Head of Model Risk
- **Detective** [planned]: Compute the population stability index of every input and of the score each month; above 0.25 the lender is alerted and the model is reviewed within 30 days.
  Recorded by: Mara Lindqvist, Head of Model Risk

3.8 Data governance

Is the provenance of all training, validation, and test datasets documented (source, collection date, license, version)? HIGH (12)

Article 10(2) requires data governance practices including documentation of dataset provenance. TraceForge can automate this.

Likelihood	likely (4)	**Severity**	moderate (3)
Residual likelihood	likely (4)	**Residual severity**	moderate (3)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.10(2), Art.10(2)(b)

NIST AI RMF: MAP 2.3 | ISO/IEC 42001: Annex A.7

Accepted: The creator, DOI, licence and file dates are recorded, but when, where and from whom the 1994 records were collected is not published and cannot be recovered. The gap is stated in the limitations section, and the model is retrained on recent lender data before release, so the remaining provenance gap is accepted.
Accepted by: Mara Lindqvist, Head of Model Risk

Are data quality checks (completeness, consistency, accuracy) applied and documented before training and fine-tuning? MEDIUM (9)

Article 10(3) requires training, validation and testing data sets to be, to the best extent possible, free of errors and complete in view of the intended purpose. Article 10(2)(c) covers data-preparation processing operations such as cleaning.

Likelihood	possible (3)	**Severity**	moderate (3)
Residual likelihood	possible (3)	**Residual severity**	moderate (3)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.10(3), Art.10(2)(c)

NIST AI RMF: MAP 2.3 | ISO/IEC 42001: Annex A.7

Is there a data retention and deletion schedule that covers both training data and inference logs? LOW (4)

Data retained beyond its purpose creates ongoing legal and privacy risk. GDPR Article 5(1)(e) requires storage limitation.

Likelihood	unlikely (2)	**Severity**	minor (2)
Residual likelihood	unlikely (2)	**Residual severity**	minor (2)
Source	question_bank	**Regulatory status**	settled

Article refs: Art.19(1), Art.26(6)

NIST AI RMF: MEASURE 2.10 | ISO/IEC 42001: Annex A.7

Knowledge gap: its test requirements are listed in section 5.

4. Residual risk summary

Risk	Inherent	Residual	Mitigations	Decision and rationale
Are model degradation scenarios (data drift, concept drift, distribution shift) monitored and have defined response proc...	**20 (critical)**	**20 (critical)**	**2**	**Open**
Does the system process personal data or special category data (health, biometric, ethnicity, religion)?	**20 (critical)**	**20 (critical)**	**1**	**Open**
Are accuracy, precision, recall, and calibration metrics documented across all relevant demographic subgroups?	16 (high)	16 (high)	1	Open
Has the system been tested against out-of-distribution inputs, edge cases, and adversarial perturbations?	16 (high)	16 (high)	1	Open
Has the training dataset been audited for representation imbalances across protected characteristics (gender, ethnicity,...	16 (high)	16 (high)	1	Open
Have demographic parity, equalised odds, or other fairness metrics been computed and documented for each protected group...	16 (high)	16 (high)	1	Open
Are system performance metrics (accuracy, error rates, latency) monitored in production with alerts for degradation?	12 (high)	12 (high)	1	Open

Risk	Inherent	Residual	Mitigations	Decision and rationale
Could the system restrict or affect access to rights protected under the EU Charter of Fundamental Rights?	12 (high)	12 (high)	1	Open
Could vulnerable populations (elderly, children, people with disabilities) be disproportionately harmed by system errors...	12 (high)	12 (high)	1	Open
Demographic bias in credit scoring outputs	12 (high)	12 (high)	1	Open
Does the system use proxy variables that may correlate with protected characteristics (e.g. postcode, school name, emplo...	20 (critical)	12 (high)	2	Open
Is an explanation of the system's decision-making logic available to affected individuals upon request?	12 (high)	12 (high)	1	Open
Is the human oversight load (volume of cases reviewed per reviewer per day) realistic for meaningful review, not rubber-...	12 (high)	12 (high)	0	Open
Is the provenance of all training, validation, and test datasets documented (source, collection date, license, version)?	12 (high)	12 (high)	0	Accepted: The creator, DOI, licence and file dates are recorded, but when, where and from whom the 1994 records were collected is not published and cannot be recovered. The gap is stated in the limitations section, and the model is retrained on recent lender data before release, so the remaining provenance gap is accepted.

Risk	Inherent	Residual	Mitigations	Decision and rationale
Is the system designed to make its reasoning interpretable to human reviewers (e.g. via explainability methods, confiden...	12 (high)	12 (high)	1	Open
Are data quality checks (completeness, consistency, accuracy) applied and documented before training and fine-tuning?	9 (medium)	9 (medium)	0	Within appetite
Are human reviewers trained on the system's capabilities, limitations, and known biases before making decisions based on...	9 (medium)	9 (medium)	0	Within appetite
Are users notified that they are interacting with or being assessed by an AI system?	9 (medium)	9 (medium)	0	Within appetite
Has a Data Protection Impact Assessment (DPIA) been carried out, with the advice of the Data Protection Officer where on...	9 (medium)	9 (medium)	0	Within appetite
Has a fundamental rights impact assessment (FRIA) been conducted prior to deployment?	9 (medium)	9 (medium)	0	Within appetite
Is there a defined performance floor (minimum acceptable metric threshold) below which the system is taken offline?	9 (medium)	9 (medium)	0	Within appetite
Lack of explanation for credit decisions	9 (medium)	9 (medium)	1	Within appetite
Are affected individuals informed of the AI system's role in decisions that affect them, in a language they understand?	6 (medium)	6 (medium)	0	Within appetite
Are bias test results documented and reviewed at each model update, not just at initial deployment?	6 (medium)	6 (medium)	0	Within appetite
Does the system documentation include clear capability limitations, known failure modes, and out-of-distribution behavio...	9 (medium)	6 (medium)	1	Within appetite

Risk	Inherent	Residual	Mitigations	Decision and rationale
Has the training data been scanned for inadvertent PII inclusion, and is PII redaction/pseudonymisation applied before t...	6 (medium)	6 (medium)	0	Within appetite
Is data minimisation enforced: does the model use only the minimum personal data necessary for the intended purpose?	12 (high)	6 (medium)	1	Within appetite
Is there a data retention and deletion schedule that covers both training data and inference logs?	4 (low)	4 (low)	0	Within appetite

5. Article 9(6) to 9(8) test requirements

Risk item	Metric	Threshold	Article
`8c42055a`	demographic_parity	`<= 0.10`	Art.9(8)
`555fecb1`	explanation_coverage	`>= 0.90`	Art.13
`60278de0`	false_negative_rate	`<= 0.01`	Art.9(2)(b)
`6c48e838`	false_negative_rate	`<= 0.01`	Art.9(2)(b)
`351bedb2`	fundamental_rights_impact_score	`= 0`	Art.9(2)(a)
`ff5260d1`	fundamental_rights_impact_score	`= 0`	Art.9(2)(a)
`f44ccbdf`	fundamental_rights_impact_score	`= 0`	Art.9(2)(a)
`f6c25aec`	demographic_parity	`<= 0.10`	Art.9(8)
`c0b71de9`	demographic_parity	`<= 0.10`	Art.9(8)
`aed4c486`	demographic_parity	`<= 0.10`	Art.9(8)
`25c401d1`	demographic_parity	`<= 0.10`	Art.9(8)
`f678a8c1`	pii_leakage_rate	`= 0.00`	Art.9(8)
`3e189772`	pii_leakage_rate	`= 0.00`	Art.9(8)
`d936c7f0`	pii_leakage_rate	`= 0.00`	Art.9(8)
`7ff6fe87`	explanation_coverage	`>= 0.90`	Art.13
`a038286b`	explanation_coverage	`>= 0.90`	Art.13
`c3ff7160`	explanation_coverage	`>= 0.90`	Art.13

Risk item	Metric	Threshold	Article
`ce393cc6`	override_latency_ms	`<= 500`	Art.14
`0265c848`	override_latency_ms	`<= 500`	Art.14
`27dd0821`	override_latency_ms	`<= 500`	Art.14
`e541ddff`	faithfulness	`>= 0.85`	Art.9(8)
`905e8cd6`	faithfulness	`>= 0.85`	Art.9(8)
`3e4bcc98`	faithfulness	`>= 0.85`	Art.9(8)
`85a548ab`	faithfulness	`>= 0.85`	Art.9(8)
`19dd45f3`	data_quality_score	`>= 0.95`	Art.10
`2a4a31b3`	data_quality_score	`>= 0.95`	Art.10
`a3f1e0de`	data_quality_score	`>= 0.95`	Art.10

6. Audit metadata

Every change to this register is recorded in a SHA-256 hash-chained audit trail stored in `.riskforge/audit.jsonl`. The JSON, PDF and Markdown exports of one register state show the same content digest. The `rmf.exported` entry of this export, recorded after the file was written, holds the content digest and the SHA-256 of this file.

Content digest (SHA-256)	`9e53abfb36a7d22a8ed445f185e3ef0e8fc484d72f3a1c170845985f8af4bb23`
RMF schema version	1.1.0
Question bank version	1.0.0
Generated by	RiskForge 2.0.0
Generated at	04 November 2026 10:52 UTC
Chain proof	53 entries of the project audit chain, seq 0 to 52
Receipt	seq 52, entry hash `5b526ceb7174932c4b53dc905af33c6ba0b0b0eb57319fcd9d93cec53bf0bdd7`
Signatures	Unsigned
Sign-offs	None recorded

Change log

Entries of the chain proof for this system and for the project.

Seq	Time (UTC)	Event	Actor
0	2026-11-04 10:51	system.created	cli
1	2026-11-04 10:51	system.classified	cli
2	2026-11-04 10:51	register.created	Mara Lindqvist
3	2026-11-04 10:51	risk_item.created	Mara Lindqvist
4	2026-11-04 10:51	risk_item.created	Mara Lindqvist
5	2026-11-04 10:51	risk_item.created	Mara Lindqvist
6	2026-11-04 10:51	risk_item.created	Mara Lindqvist
7	2026-11-04 10:51	risk_item.created	Mara Lindqvist
8	2026-11-04 10:51	risk_item.created	Mara Lindqvist
9	2026-11-04 10:51	risk_item.created	Mara Lindqvist
10	2026-11-04 10:51	risk_item.created	Mara Lindqvist
11	2026-11-04 10:51	risk_item.created	Mara Lindqvist
12	2026-11-04 10:51	risk_item.created	Mara Lindqvist
13	2026-11-04 10:51	risk_item.created	Mara Lindqvist
14	2026-11-04 10:51	risk_item.created	Mara Lindqvist
15	2026-11-04 10:51	risk_item.created	Mara Lindqvist
16	2026-11-04 10:51	risk_item.created	Mara Lindqvist
17	2026-11-04 10:51	risk_item.created	Mara Lindqvist
18	2026-11-04 10:51	risk_item.created	Mara Lindqvist
19	2026-11-04 10:51	risk_item.created	Mara Lindqvist
20	2026-11-04 10:51	risk_item.created	Mara Lindqvist
21	2026-11-04 10:51	risk_item.created	Mara Lindqvist
22	2026-11-04 10:51	risk_item.created	Mara Lindqvist
23	2026-11-04 10:51	risk_item.created	Mara Lindqvist
24	2026-11-04 10:51	risk_item.created	Mara Lindqvist
25	2026-11-04 10:51	risk_item.created	Mara Lindqvist
26	2026-11-04 10:51	risk_item.created	Mara Lindqvist
27	2026-11-04 10:51	risk_item.created	Mara Lindqvist
28	2026-11-04 10:51	risk_item.created	Mara Lindqvist
29	2026-11-04 10:51	risk_item.created	Mara Lindqvist

Seq	Time (UTC)	Event	Actor
30	2026-11-04 10:51	risk_item.created	Mara Lindqvist
31	2026-11-04 10:51	mitigation.added	Mara Lindqvist, Head of Model Risk
32	2026-11-04 10:51	mitigation.added	Mara Lindqvist, Head of Model Risk
33	2026-11-04 10:51	mitigation.added	Mara Lindqvist, Head of Model Risk
34	2026-11-04 10:51	mitigation.added	Mara Lindqvist, Head of Model Risk
35	2026-11-04 10:51	mitigation.added	Mara Lindqvist, Head of Model Risk
36	2026-11-04 10:51	mitigation.added	Mara Lindqvist, Head of Model Risk
37	2026-11-04 10:51	mitigation.added	Mara Lindqvist, Head of Model Risk
38	2026-11-04 10:51	mitigation.added	Mara Lindqvist, Head of Model Risk
39	2026-11-04 10:51	mitigation.added	Mara Lindqvist, Head of Model Risk
40	2026-11-04 10:51	mitigation.added	Mara Lindqvist, Head of Model Risk
41	2026-11-04 10:51	mitigation.added	Mara Lindqvist, Head of Model Risk
42	2026-11-04 10:51	mitigation.added	Mara Lindqvist, Head of Model Risk
43	2026-11-04 10:51	mitigation.added	Mara Lindqvist, Head of Model Risk
44	2026-11-04 10:51	mitigation.added	Mara Lindqvist, Head of Model Risk
45	2026-11-04 10:51	mitigation.added	Mara Lindqvist, Head of Model Risk
46	2026-11-04 10:51	mitigation.added	Mara Lindqvist, Head of Model Risk
47	2026-11-04 10:51	mitigation.added	Mara Lindqvist, Head of Model Risk
48	2026-11-04 10:51	mitigation.added	Mara Lindqvist, Head of Model Risk
49	2026-11-04 10:51	risk.accepted	Mara Lindqvist, Head of Model Risk
50	2026-11-04 10:51	tests.generated	cli
51	2026-11-04 10:52	rmf.exported	cli
52	2026-11-04 10:52	rmf.exported	cli

Verify integrity at any time: `riskforge verify`. Check the JSON export of this file with `riskforge verify --file`.